Did Liquidity Govern Stablecoin Peg Pass-Through? Evidence from the March 2023 USDC Depeg*

Samson Lam

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Abstract

This study asks whether liquidity governed how stablecoin peg stress passed through to Bitcoin's price during the March 2023 USDC depeg. Using 1-minute data on BTC/USD, BTC/USDT, BTC/USDC, USDT/USD and USDC/USD, we decompose each stablecoin's cross-currency basis into its peg deviation and a triangular arbitrage deviation. We then model the arbitrage deviation's response to the peg as a function of lagged illiquidity. During the crisis, the peg deviation carried most of the basis variance, while the triangular arbitrage deviation decayed with a half-life below 1 minute, compared with about 1 hour for the USDC peg and over 2.5 hours for the USDT peg. The results show that liquidity did not measurably govern the pass-through. The effect of USDC/USD illiquidity is significant at 1-minute sampling but not robust. The stress that reached Bitcoin's price was the peg itself, not a failure of arbitrage.[1]

1 Introduction

A stablecoin is a cryptocurrency that trades at a fixed target against a fiat currency, one US dollar for Tether (USDT) and USD Coin (USDC). Exchanges quote Bitcoin against both the dollar and stablecoins, so the stablecoin price of Bitcoin is a dollar price only while the peg holds. Empirically, stablecoin pegs do fail, and the failures spill over to other crypto-assets. Perez Riaza and Gnabo [15] find that a Tether depeg raises the probability of a jump in BTC/USD almost fivefold within 5 minutes, Foley et al. [8] find that downward Tether depegs lower the returns of major cryptocurrencies, and Bartůněk [3] finds that depegging raises the realized volatility of Bitcoin.

When a peg breaks, triangular arbitrage ties BTC/USDC to BTC/USD and USDC/USD, so BTC/USDC should move away from BTC/USD by the peg deviation. However, arbitrage is not free. Makarov and Schoar [14] document large arbitrage deviations across cryptocurrency exchanges, Reynolds et al. [18] find deviations from triangular arbitrage parity in Bitcoin markets but not among fiat currencies, and Lyons and Viswanath-Natraj [13] show that Tether's peg deviations halved once arbitrage with the issuer became easier. If thin order books limit triangular arbitrage, a depeg should pass slowly and incompletely into the stablecoin price of Bitcoin when the peg moves most. This study asks whether liquidity governed how stablecoin peg stress passed into the stablecoin price of Bitcoin.

We address this question using the USDC depeg in March 2023. Silicon Valley Bank, which held a substantial part of the reserves of USDC [15], failed on 10 March 2023 [4]. USDC traded as low as 0.863 dollars on 11 March, before the joint statement of 12 March protected all depositors of the bank [20]. We decompose each stablecoin's cross-currency basis into its peg deviation and a triangular arbitrage deviation. We then model the response of the triangular arbitrage deviation to the peg as a function of lagged illiquidity.

We find that liquidity did not govern the pass-through of the stablecoin peg. During the crisis, the peg deviation carried most of the basis variance, and the triangular arbitrage deviation decayed with a half-life below 1 minute, compared with about 1 hour for the USDC peg and over 2.5 hours for the USDT peg. Arbitrage between the three pairs therefore kept working even when the

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[1]Code and replication data are available at *GitHub*.

USDC pairs were thin, and the stress on Bitcoin's stablecoin price came from the peg itself. The estimated effect of USDC/USD illiquidity is significant at 1% at 1-minute sampling, but only at 10% at 5-minute sampling.

Section 2 describes the data and the depeg events. Section 3 states the methodology, and Section 4 reports the results. Section 5 concludes with the implications for stablecoin regulation.

2 Data

2.1 Sample and notation

We use 1-minute data from Binance.US for five pairs, {BTC/USD, BTC/USDT, BTC/USDC, USDT/USD and USDC/USD}, from 1 to 21 March 2023. Each pair includes the high, low, close, base volume and quote volume. Write $P_t^{X/Y}$ for the close price of one unit of X in units of Y at time t, and $Q \in \{\text{USDT}, \text{USDC}\}$ for a stablecoin. Deviations are in basis points (bps), so log prices are scaled by $m = 10^4$.

A minute without a trade has zero volume and carries a stale price, which is the non-synchronous trading problem of Lo and MacKinlay [12]. Since a deviation constructed from three closing prices is meaningful only when all three are current, we define F_t as an indicator that all three pairs of the triangle trade at time t, and FF_t as an indicator that both t and $t-1$ have $F = 1$,

$$F_t = \mathbf{1}\{v_t^{\text{BTC/USD}} > 0,\ v_t^{\text{BTC}/Q} > 0,\ v_t^{Q/\text{USD}} > 0\}, \qquad FF_t = F_t F_{t-1}, \tag{1}$$

where v is the pair volume. This is a stricter form of the refresh-time sampling of Barndorff-Nielsen et al. [2], which samples once every asset has traded.

2.2 Depeg event identification

Following Perez Riaza and Gnabo [15], we sample the Q/USD close every 5 minutes, π_k. A bar is a breach when $|\pi_k - 1| \geq \tau$, with $\tau = 0.5\%$ in the baseline [15, 17]. An event starts at the last bar before the breach when the price equals 1 dollar exactly, or at the first breach bar when the previous bar is at the peg. This adapts the retrospective start of Perez Riaza and Gnabo [15] in their Section 6.1. We merge breaches at most 20 minutes apart and drop events shorter than 60 minutes.

Estimates for one stablecoin use its own window, and estimates that use all pairs use the union of the two windows, which equals the USDT window. Table 1 reports the windows, including two alternative windows. The baseline windows run from 21:10 on 10 March to 13:10 on 15 March for USDT and from 01:20 on 11 March to 02:30 on 14 March for USDC. Minutes before a window form the pre-crisis regime, minutes inside it the crisis regime, and minutes after it the post-crisis regime.

Stablecoin	Threshold and start	Start	End	Hours
USDT	±0.5%, last-peg start (baseline)	10 Mar 21:10	15 Mar 13:10	112.0
USDT	±1.0%, last-peg start (R3)	10 Mar 21:10	12 Mar 22:40	49.5
USDT	±0.5%, crossing start (R6)	11 Mar 00:55	14 Mar 11:15	82.3
USDC	±0.5%, last-peg start (baseline)	11 Mar 01:20	14 Mar 02:30	73.2
USDC	±1.0%, last-peg start (R3)	11 Mar 01:20	13 Mar 18:45	65.4
USDC	±0.5%, crossing start (R6)	11 Mar 04:00	13 Mar 18:55	62.9

Table 1: Depeg windows from the 5-minute Q/USD close. Times are UTC in March 2023.

2.3 Liquidity measures

We measure illiquidity with the Kyle estimator of Kyle and Obizhaeva [11], which Brauneis et al. [5] find, together with the Amihud ratio, best tracks the level of liquidity in cryptocurrency markets. For a pair with 1-minute log return r_t and quote volume V_t,

$$\mathrm{Kyle}_t = m \left(\frac{\frac{1}{5}\sum_{i=0}^{4} r_{t-i}^2}{\sum_{i=0}^{4} V_{t-i}} \right)^{1/3}, \tag{2}$$

so that Kyle_t rises with volatility per unit of volume. The Corwin–Schultz spread estimator of Corwin and Schultz [6] (CS), which Brauneis et al. [5] find, together with that of Abdi and Ranaldo [1], best tracks time variation in spreads, enters the pass-through regression with the realized volatility (RV). Appendix A defines both.

Table 2 shows that the crisis affected the USDC pairs most. The Kyle illiquidity of USDC/USD rose from 1.23 to 7.23 and that of BTC/USDC from 3.08 to 6.37, compared with a rise from 0.79 to 1.02 for BTC/USD. Trading in USDC/USD also rose from 1.13 to 9.76 million dollars per day.

	Kyle			Quote volume per day		
Pair	pre	crisis	post	pre	crisis	post
BTC/USD	0.79	1.02	0.88	141	254	333
BTC/USDT	1.14	1.40	1.73	55.4	113	58.1
BTC/USDC	3.08	6.37	5.33	4.15	5.38	3.71
USDT/USD	0.91	1.48	1.44	3.21	8.99	3.89
USDC/USD	1.23	7.23	3.79	1.13	9.76	0.94

Table 2: Liquidity by pair and regime over the union window. Kyle is the illiquidity of Eq. (2), and quote volume is in millions of the quote currency.

3 Methodology

3.1 Basis decomposition

For stablecoin Q, we decompose the cross-currency basis D_t into the peg deviation δ_t and the triangular arbitrage deviation b_t,

$$D_t = m\left(\ln P_t^{\mathrm{BTC}/Q} - \ln P_t^{\mathrm{BTC/USD}}\right), \qquad \delta_t = m \ln P_t^{Q/\mathrm{USD}}, \qquad b_t = -D_t - \delta_t. \tag{3}$$

Triangular arbitrage parity, $P^{\mathrm{BTC/USD}} = P^{\mathrm{BTC}/Q} P^{Q/\mathrm{USD}}$, implies $b_t = 0$ [18]. Full pass-through of the peg therefore means $D_t = -\delta_t$, and b_t is the part of the basis that the peg does not explain. The variance of the basis decomposes into

$$\mathrm{Var}(D) = \mathrm{Var}(\delta) + \mathrm{Var}(b) + 2\,\mathrm{Cov}(\delta, b). \tag{4}$$

3.2 Half-life of deviations

For $x \in \{\delta, b\}$, we estimate a first-order autoregressive (AR(1)) model and the half-life HL in minutes [13],

$$x_t = c + \rho x_{t-1} + u_t, \qquad \mathrm{HL} = \frac{\ln 0.5}{\ln \rho}. \tag{5}$$

3.3 Pass-through regression

The pass-through regression is

$$\Delta b_t = \alpha + \beta b_{t-1} + \left(\lambda_0 + \boldsymbol{\lambda}_1^\top \tilde{\mathbf{z}}_{t-1}\right) \delta_{t-1} + \boldsymbol{\theta}^\top \tilde{\mathbf{z}}_{t-1} + \kappa \widetilde{\mathrm{RV}}_{t-1} + \varepsilon_t, \tag{6}$$

where the vector $\mathbf{z}_{t-1}$ holds the lagged CS and Kyle illiquidity of BTC/Q and the Kyle illiquidity of Q/USD, and RV_{t-1} is the lagged RV of BTC/USD. A tilde marks standardization within the estimation sample.

The coefficient β measures the reversion speed of b, and λ_0 measures how the lagged peg deviation moves b at average liquidity. The vector $\boldsymbol{\lambda}_1$ measures how that response changes with a 1 standard deviation rise in each illiquidity measure, so liquidity governs the pass-through when $\boldsymbol{\lambda}_1 \neq \mathbf{0}$.

3.4 Spillover index and Granger causality

We model the joint dynamics of the five pairs by a vector autoregression (VAR) on 5-minute log returns scaled by m,

$$\mathbf{r}_t = \mathbf{c} + \sum_{l=1}^{p} \boldsymbol{\Phi}_l \mathbf{r}_{t-l} + \boldsymbol{\varepsilon}_t, \qquad \mathrm{E}\left[\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t^\top\right] = \boldsymbol{\Sigma}, \tag{7}$$

estimated in each regime of the union window with the lag order p chosen by the Bayesian information criterion (BIC) up to 6 lags. The generalized forecast error variance decomposition of Koop et al. [10] and Pesaran and Shin [16] gives the share of the H-step forecast error variance of pair i due to shocks in pair j,

$$\phi_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} \left(\mathbf{e}_i^\top \mathbf{A}_h \boldsymbol{\Sigma} \mathbf{e}_j\right)^2}{\sum_{h=0}^{H-1} \mathbf{e}_i^\top \mathbf{A}_h \boldsymbol{\Sigma} \mathbf{A}_h^\top \mathbf{e}_i}, \qquad \tilde{\phi}_{ij} = \frac{100\, \phi_{ij}}{\sum_l \phi_{il}}, \tag{8}$$

where $\mathbf{A}_h$ are the moving-average coefficient matrices, σ_{jj} is the jth diagonal element of $\boldsymbol{\Sigma}$, and $\mathbf{e}_i$ selects pair i. We set $H = 10$ (50 minutes), and the total spillover index of Diebold and Yilmaz [7] is

$$S = \frac{1}{5} \sum_{i \neq j} \tilde{\phi}_{ij}. \tag{9}$$

Pair j Granger-causes pair i [9] when the lags of j help predict i in Eq. (7), which we test with a Wald test that all p coefficients of j in the equation of i are zero. Appendix C states the full formulas and reports all directed tests.

4 Empirical results

4.1 Decomposition of the basis

Figure 1 plots the time series around each crisis window. Before the depeg, all three series stayed close to zero. Once the peg broke, the basis D mirrored the peg deviation δ almost exactly, while the triangular arbitrage deviation b fluctuated around zero with short spikes.

Table 3 confirms these findings. During the crisis, the correlation of D with $-\delta$ is 0.99 for both stablecoins, and $\mathrm{Var}(\delta)/\mathrm{Var}(D)$ is 1.02 for USDT and 1.00 for USDC, so the peg deviation accounts for almost the entire variance of the basis. Before the crisis, the same correlation was only 0.59 for USDT and 0.04 for USDC, primarily because the pegs barely moved and the basis consisted mostly of noise in b. In dollar terms, BTC/USDC and BTC/USDT therefore traded at the BTC/USD

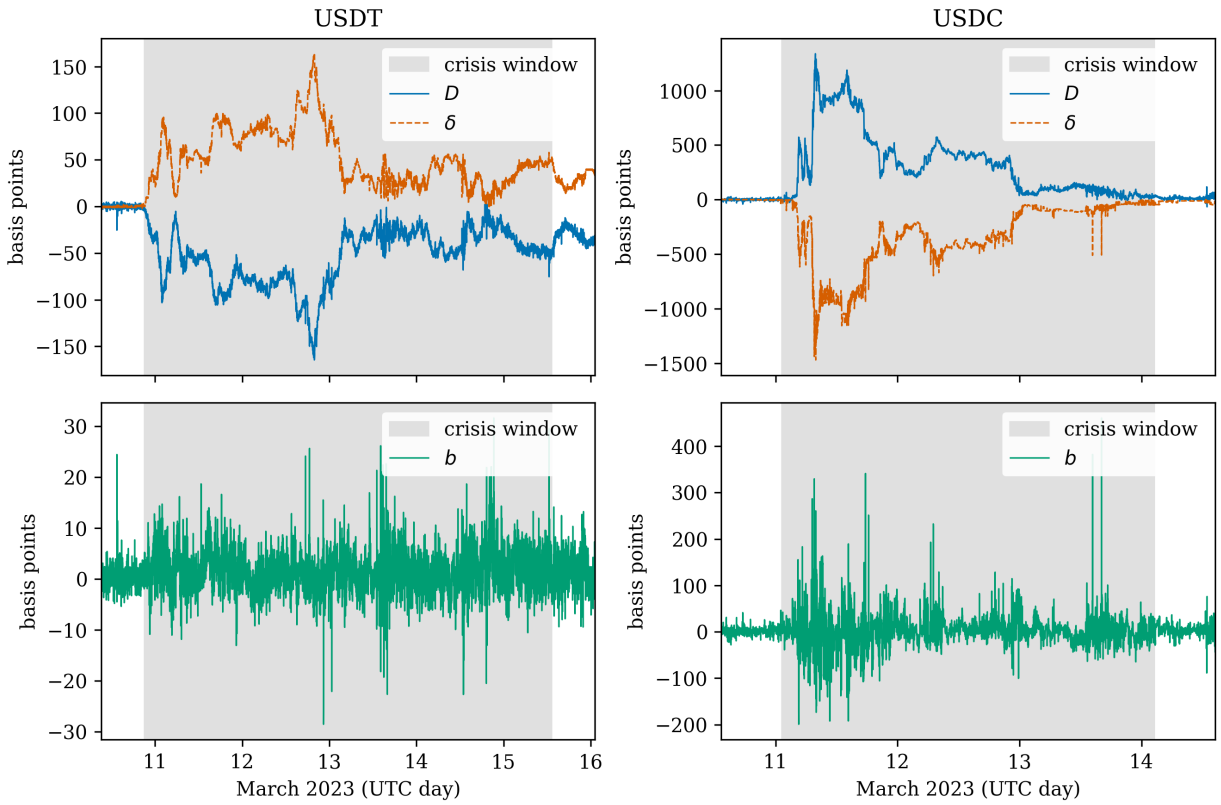


Figure 1: Cross-currency basis D and peg deviation δ (top), and triangular arbitrage deviation b (bottom) in bps, 1-minute close prices from 12 hours before to 12 hours after each crisis window (shaded), UTC.

	USDT			USDC		
	pre	crisis	post	pre	crisis	post
Mean D	−0.11	−53.2	−26.0	0.13	399	7.23
Mean δ	0.12	51.6	25.3	−0.61	−400	−7.17
Var(δ)/Var(D)	0.56	1.02	0.97	0.67	1.00	0.63
Mean b	−0.01	1.57	0.70	0.47	1.72	−0.06
$\lvert b\rvert$ median	1.03	2.50	2.39	2.23	13.4	5.10
$\lvert b\rvert$ 99th percentile	5.37	12.3	10.9	17.3	140	26.5

Table 3: Decomposition of Eq. (3) on synchronous minutes ($F_t = 1$), in bps. Standard deviations and the remaining variance terms are in Table B1.

price throughout the crisis, and the gap in their quoted prices reflected the stablecoin's value rather than a separate Bitcoin price.

The crisis widened the distribution of b without moving its mean. The mean of b is below 2 bps (1.57 for USDT and 1.72 for USDC), while for USDC the median of $|b|$ rose from 2.23 to 13.4 bps. Triangular arbitrage deviations thus became larger and more frequent but did not persist in one direction, as expected from transient trading frictions rather than a sustained mispricing.

4.2 Half-lives

Table 4 reports the half-lives. During the crisis, the half-life of b is 0.82 minutes for USDT and 0.52 minutes for USDC, compared with 158 and 64 minutes for δ. Before the crisis, both were short, with peg half-lives of 5.7 and 0.9 minutes, since the pegs then only fluctuated around 1 dollar. Arbitrage therefore closed triangular deviations within a minute even at the height of the stress, while the peg deviation persisted for hours. The persistence of the basis was hence the persistence of the peg.

		pre	crisis	post
USDT	δ	5.72	158	27.8
	b	0.45	0.82	0.64
USDC	δ	0.92	63.9	2.47
	b	0.43	0.52	0.61

Table 4: Half-life in minutes from Eq. (5) on consecutive synchronous minutes ($FF_t = 1$).

4.3 Liquidity and pass-through

Table 5 reports the pass-through regression. The coefficient β is between -0.59 and -0.90 and significant at 1% in every regime. Reversion was slower during the crisis than before, at 59% versus 81% for USDT and 74% versus 90% for USDC, but it remained fast.

	USDT			USDC		
	pre	crisis	post	pre	crisis	post
β	-0.810^{***}	-0.589^{***}	-0.688^{***}	-0.898^{***}	-0.744^{***}	-0.691^{***}
	(-13.77)	(-18.92)	(-23.47)	(-13.42)	(-12.84)	(-7.88)
λ_0	-0.0782^{***}	-0.0098^{***}	-0.0387^{**}	-0.7004^{***}	0.0015	0.0780
	(-2.82)	(-2.88)	(-2.53)	(-7.67)	(0.24)	(1.26)
λ_1, CS of BTC/Q	0.0161	-0.0019	-0.0096^{**}	0.2642^{***}	-0.0006	-0.0079
	(0.61)	(-1.02)	(-2.09)	(2.63)	(-0.13)	(-0.24)
λ_1, Kyle of BTC/Q	0.0124	-0.0045^{*}	0.0205^{**}	0.0735	0.0027	-0.0220
	(0.34)	(-1.65)	(2.20)	(1.08)	(1.06)	(-1.25)
λ_1, Kyle of Q/USD	0.0301^{*}	-0.0009	-0.0060	0.1195^{***}	0.0137^{***}	-0.0317
	(1.89)	(-0.31)	(-1.01)	(7.52)	(2.91)	(-1.06)
Joint p, $\lambda_1 = 0$	0.208	0.263	0.021	<0.001	0.007	0.568

Table 5: Pass-through regression of Eq. (6) on consecutive synchronous minutes, with t statistics in parentheses. *, ** and *** denote significance at 10%, 5% and 1%.

For USDT during the crisis, λ_0 is -0.0098 and statistically significant. At the mean premium of 51.6 bps, this implies a move in b of about -0.51 bps per minute. The liquidity terms are jointly insignificant, so liquidity did not change this response.

For USDC during the crisis, λ_0 is 0.0015 but not statistically significant, so at average liquidity the peg deviation did not predict the next change in b. The coefficient on the Kyle illiquidity of USDC/USD is 0.0137, and the liquidity terms are jointly significant. At the mean discount of -400 bps, a 1 standard deviation rise in this illiquidity implies a move in b of about -5.5 bps per minute. The CS and Kyle illiquidity of BTC/USDC are insignificant, so the effect runs through the USDC/USD pair, in which the peg is set, rather than through the Bitcoin pair. When USDC/USD was illiquid, the peg therefore passed into BTC/USDC with a small delay. The effect is small relative to the discount and, as Section 4.5 shows, not robust.

Outside the crisis, the liquidity terms are also jointly significant for USDC before the crisis and for USDT after it.

4.4 Spillovers and lead-lag relations

The BIC selects 6, 2 and 4 lags in the three regimes, and the total spillover index barely moved, at 39.7, 37.5 and 39.4. The crisis did not change overall connectedness but did change its composition. Table 6 shows that the share of BTC/USDC forecast error variance due to BTC/USD fell from 32.4% before the crisis to 21.4% during it and returned to 31.1% after it. USDC/USD took part of its place, explaining 5.9% of the forecast error variance of BTC/USDC during the crisis. Meanwhile, BTC/USDC explained 10.3% of the forecast error variance of USDC/USD.

The Granger tests point to the same conclusion. BTC/USD led BTC/USDC in every regime and never the reverse, while USDC/USD and BTC/USDC led each other only during the crisis. In the USDT triangle, BTC/USDT led USDT/USD in every regime. Price discovery for Bitcoin therefore stayed in BTC/USD throughout, while during the crisis BTC/USDC partly became a market for USDC itself, so that its prices and the USDC peg predicted each other.

	pre	crisis	post
Panel A. Variance shares (%)			
Share of BTC/USDC from BTC/USD	32.4	21.4	31.1
Share of BTC/USDC from USDC/USD	0.0	5.9	0.2
Share of USDC/USD from BTC/USDC	0.2	10.3	0.3
Panel B. Granger causality, Wald statistic			
BTC/USD $\rightarrow$ BTC/USDC	17.79***	10.11***	28.77***
BTC/USDC $\rightarrow$ BTC/USD	3.51	1.89	7.54
USDC/USD $\rightarrow$ BTC/USDC	1.37	14.87***	5.27
BTC/USDC $\rightarrow$ USDC/USD	6.00	114.19***	7.06
BTC/USD $\rightarrow$ BTC/USDT	16.76**	6.99**	29.54***
BTC/USDT $\rightarrow$ BTC/USD	10.06	13.57***	3.05
BTC/USDT $\rightarrow$ USDT/USD	28.45***	15.22***	28.42***
USDT/USD $\rightarrow$ BTC/USDT	5.15	0.03	4.84

Table 6: Spillovers of 5-minute returns over the union window. Shares are the normalized $\tilde{\phi}_{ij}$ of Eq. (8) with $H = 10$. *, ** and *** denote significance at 10%, 5% and 1%. The full matrices and all Granger tests are in Appendix C.

4.5 Robustness testing

Table 7 repeats the key estimates under different variants. R1 re-estimates Tables 4 and 5 on 5-minute bars, and R2 enters each liquidity measure alone. The effect of USDC/USD illiquidity

is significant at 1-minute sampling but not robust. Its coefficient stays similar, but at 5-minute sampling it is significant only at 10%, and the liquidity terms are jointly insignificant, with a joint p of 0.106. The effect is too weak to survive at different sampling frequencies.

	USDC pass-through			Half-life, USDT / USDC		
Check	λ_1	Joint p	USDT λ_0	b	δ	Share
R0 baseline	0.0137^{***} (2.91)	0.007	-0.0098^{***} (-2.88)	0.82 / 0.52	158 / 63.9	21.4
R1 5-minute bars	0.0143^{*} (1.83)	0.106	-0.0132^{***} (-3.16)	2.47 / 2.27	284 / 219	–
R2 CS of BTC/USDC	0.0005 (0.11)	0.910	-0.0084^{**} (-2.36)	–	–	–
R2 Kyle of BTC/USDC	0.0040 (1.33)	0.185	-0.0091^{**} (-2.56)	–	–	–
R2 Kyle of USDC/USD	0.0149^{***} (2.93)	0.003	-0.0093^{***} (-2.70)	–	–	–

Table 7: Key crisis estimates, with t statistics in parentheses. Share is the percentage of BTC/USDC forecast error variance due to BTC/USD. *, ** and *** denote significance at 10%, 5% and 1%.

5 Conclusion

In this study, we asked whether liquidity governed how stablecoin peg stress passed through to Bitcoin's stablecoin price during the March 2023 USDC depeg. We decomposed the cross-currency basis of USDT and USDC on 1-minute data into peg deviation and triangular arbitrage deviation, estimated their half-lives, and examined whether the triangular arbitrage deviation's response to the peg depended on lagged illiquidity.

During the crisis, the peg deviation carried almost the entire variance of the basis, and the triangular arbitrage deviation closed with a half-life below 1 minute, compared with about 1 hour for the USDC peg and over 2.5 hours for the USDT peg. Liquidity did not measurably govern the pass-through. The effect of USDC/USD illiquidity is significant at 1-minute sampling but not robust. Arbitrage relationships were maintained through the crisis, and the stress that reached the stablecoin price of Bitcoin was the peg itself. Price discovery stayed in BTC/USD, while BTC/USDC partly decoupled from it and traded with the USDC peg.

The results also bear on the GENIUS Act of 2025, which brings payment stablecoins under a federal framework and requires one-to-one reserve backing and public disclosure of reserves [19]. Rules that make reserves credible and redemption reliable act on the layer that carried the stress.

The study covers only one exchange and one episode, so the results may not extend to venues with deeper USDC books or to slower depegs. The liquidity measures are approximate estimates from 1-minute bars rather than direct observations from the order book. Future work could use order book depth across several exchanges and separate the persistent peg from microstructure noise.

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A Liquidity and volatility measures

Write $\overline{P}_t$ and $\underline{P}_t$ for the high and the low of a bar. The Corwin–Schultz spread estimator of Corwin and Schultz [6] over bars $t-1$ and t and the realized volatility over the last 120 bars are

$$\psi_t = \sum_{i=0}^{1} \left(\ln \frac{\overline{P}_{t-i}}{\underline{P}_{t-i}} \right)^2, \qquad \omega_t = \left(\ln \frac{\max(\overline{P}_t, \overline{P}_{t-1})}{\min(\underline{P}_t, \underline{P}_{t-1})} \right)^2, \qquad \zeta_t = \frac{\sqrt{2\psi_t} - \sqrt{\psi_t}}{3 - 2\sqrt{2}} - \sqrt{\frac{\omega_t}{3 - 2\sqrt{2}}},$$

$$\mathrm{CS}_t = m \max\left\{ 0, \frac{2(e^{\zeta_t} - 1)}{1 + e^{\zeta_t}} \right\},$$

$$\mathrm{RV}_t = m \left(\sum_{i=0}^{119} r_{t-i}^2 \right)^{1/2}.$$

B Additional estimates

	USDT			USDC		
	pre	crisis	post	pre	crisis	post
SD D	2.17	30.4	8.46	4.12	327	10.4
SD δ	1.63	30.7	8.31	3.36	327	8.28
SD b	1.80	4.08	3.78	5.21	38.3	8.83
$\mathrm{Var}(b)/\mathrm{Var}(D)$	0.68	0.02	0.20	1.60	0.01	0.72
$2\mathrm{Cov}(\delta, b)/\mathrm{Var}(D)$	−0.24	−0.04	−0.17	−1.27	−0.01	−0.35
$\lvert b \rvert$ 95th percentile	3.32	8.47	7.72	9.76	81.0	17.3

Table B1: Remaining decomposition moments on synchronous minutes ($F_t = 1$), in bps.

	USDT			USDC		
	pre	crisis	post	pre	crisis	post
θ, CS of BTC/Q	0.001	−0.011	0.273**	0.041	−1.895	−0.695
	(0.05)	(−0.11)	(2.01)	(0.40)	(−1.01)	(−1.05)
θ, Kyle of BTC/Q	0.025	0.315*	−0.537**	−0.033	0.366	−0.502*
	(0.70)	(1.79)	(−2.07)	(−0.26)	(0.25)	(−1.69)
θ, Kyle of Q/USD	0.016	−0.106	−0.075	−0.220	5.517***	−0.081
	(0.58)	(−0.70)	(−0.44)	(−1.48)	(2.86)	(−0.14)
κ	0.158**	−0.056	0.075	0.298*	0.989	1.442***
	(2.54)	(−0.44)	(0.76)	(1.79)	(0.86)	(2.76)
R^2	0.408	0.290	0.342	0.442	0.373	0.378

Table B2: Control coefficients and fit of Eq. (6), with t statistics in parentheses. *, ** and *** denote significance at 10%, 5% and 1%.

C Spillover index and Granger causality

Write the VAR of Eq. (7) in its moving-average form,

$$\mathbf{r}_t = \boldsymbol{\mu} + \sum_{h=0}^{\infty} \mathbf{A}_h \boldsymbol{\varepsilon}_{t-h}, \qquad \mathbf{A}_0 = \mathbf{I}, \qquad \mathbf{A}_h = \sum_{l=1}^{\min(h,p)} \boldsymbol{\Phi}_l \mathbf{A}_{h-l}. \tag{10}$$

The generalized forecast error variance decomposition of Eq. (8) uses these $\mathbf{A}_h$ only [16]. Since the generalized shares of a row need not sum to 1, each row is normalized, so that $\sum_j \tilde{\phi}_{ij} = 100$ and $\sum_{i,j} \tilde{\phi}_{ij} = 500$. The total spillover index of Eq. (9) is the average share of forecast error variance that comes from the other four pairs [7].

For Granger causality from pair j to pair i, let $\boldsymbol{\gamma}_{ij} = ([\boldsymbol{\Phi}_1]_{ij}, \ldots, [\boldsymbol{\Phi}_p]_{ij})^\top$ collect the p coefficients of j in the equation of i. The null hypothesis $H_0 \colon \boldsymbol{\gamma}_{ij} = \mathbf{0}$ is tested with

$$W_{ij} = \hat{\boldsymbol{\gamma}}_{ij}^\top \left[\widehat{\mathrm{Var}} \left(\hat{\boldsymbol{\gamma}}_{ij} \right) \right]^{-1} \hat{\boldsymbol{\gamma}}_{ij} \;\sim\; \chi^2_p, \tag{11}$$

where $\widehat{\mathrm{Var}}(\hat{\boldsymbol{\gamma}}_{ij})$ is the corresponding block of the estimated covariance of the VAR coefficients [9].

Row i, from column j	BTC/USD	BTC/USDT	BTC/USDC	USDT/USD	USDC/USD
Pre-crisis					
BTC/USD	34.4	34.0	31.5	0.1	0.0
BTC/USDT	34.0	34.4	31.5	0.1	0.0
BTC/USDC	32.4	32.4	35.1	0.1	0.0
USDT/USD	0.4	0.4	0.5	98.3	0.3
USDC/USD	0.3	0.2	0.2	0.1	99.2
Crisis					
BTC/USD	41.6	40.6	17.4	0.1	0.3
BTC/USDT	40.4	41.6	17.1	0.8	0.2
BTC/USDC	21.4	21.1	51.4	0.2	5.9
USDT/USD	2.8	4.3	1.1	90.5	1.3
USDC/USD	0.5	0.4	10.3	1.3	87.5
Post-crisis					
BTC/USD	35.5	34.6	29.5	0.2	0.1
BTC/USDT	34.6	35.7	29.3	0.2	0.1
BTC/USDC	31.1	30.9	37.6	0.2	0.2
USDT/USD	1.5	1.6	1.1	95.6	0.2
USDC/USD	0.1	0.2	0.3	0.8	98.5

Table C1: Normalized generalized variance shares $\tilde{\phi}_{ij}$ in percent, row pair i from column pair j, 5-minute returns, $H = 10$.

Cause → effect	pre	crisis	post
BTC/USD → BTC/USDT	16.76**	6.99**	29.54***
BTC/USD → BTC/USDC	17.79***	10.11***	28.77***
BTC/USD → USDT/USD	24.22***	22.33***	35.03***
BTC/USD → USDC/USD	9.51	6.38**	5.59
BTC/USDT → BTC/USD	10.06	13.57***	3.05
BTC/USDT → BTC/USDC	9.23	10.92***	2.39
BTC/USDT → USDT/USD	28.45***	15.22***	28.42***
BTC/USDT → USDC/USD	4.74	1.07	8.15*
BTC/USDC → BTC/USD	3.51	1.89	7.54
BTC/USDC → BTC/USDT	3.03	1.77	8.44*
BTC/USDC → USDT/USD	4.06	0.78	3.01
BTC/USDC → USDC/USD	6.00	114.19***	7.06
USDT/USD → BTC/USD	5.21	2.83	8.45*
USDT/USD → BTC/USDT	5.15	0.03	4.84
USDT/USD → BTC/USDC	6.76	9.28***	4.87
USDT/USD → USDC/USD	1.11	5.68*	18.88***
USDC/USD → BTC/USD	2.46	0.93	6.16
USDC/USD → BTC/USDT	2.64	1.30	5.90
USDC/USD → BTC/USDC	1.37	14.87***	5.27
USDC/USD → USDT/USD	8.75	1.98	3.80

Table C2: Granger causality Wald statistics W_{ij}, 5-minute returns, with χ^2 degrees of freedom equal to the lag order (6, 2 and 4). *, ** and *** denote significance at 10%, 5% and 1%.